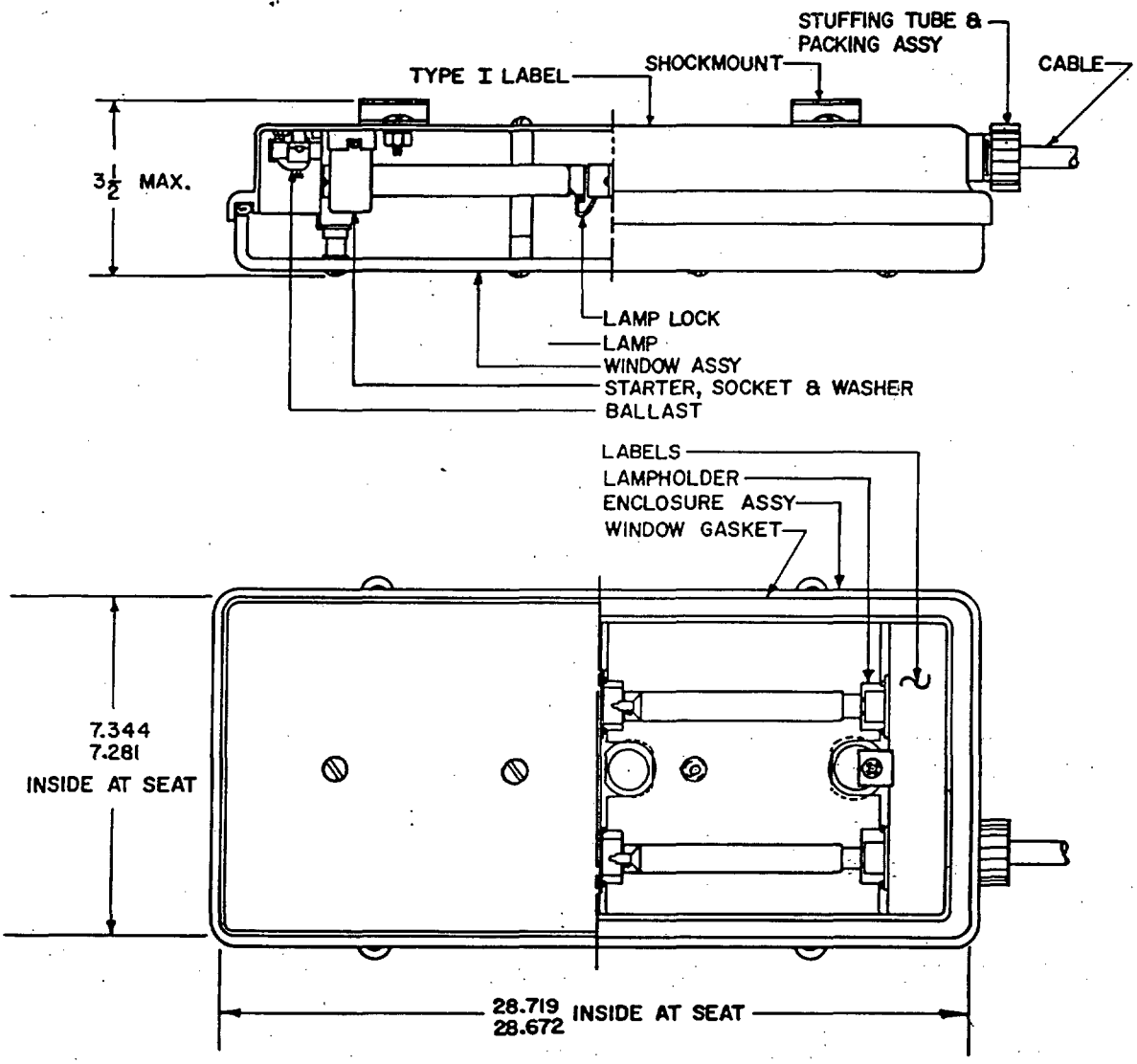
MILITARY SPECIFICATION SHEET

# FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, DIRECTIONAL

## 32 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 145.4 AND 145.5

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).




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FIGURE 1. Typical view of Symbol 145.4.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.



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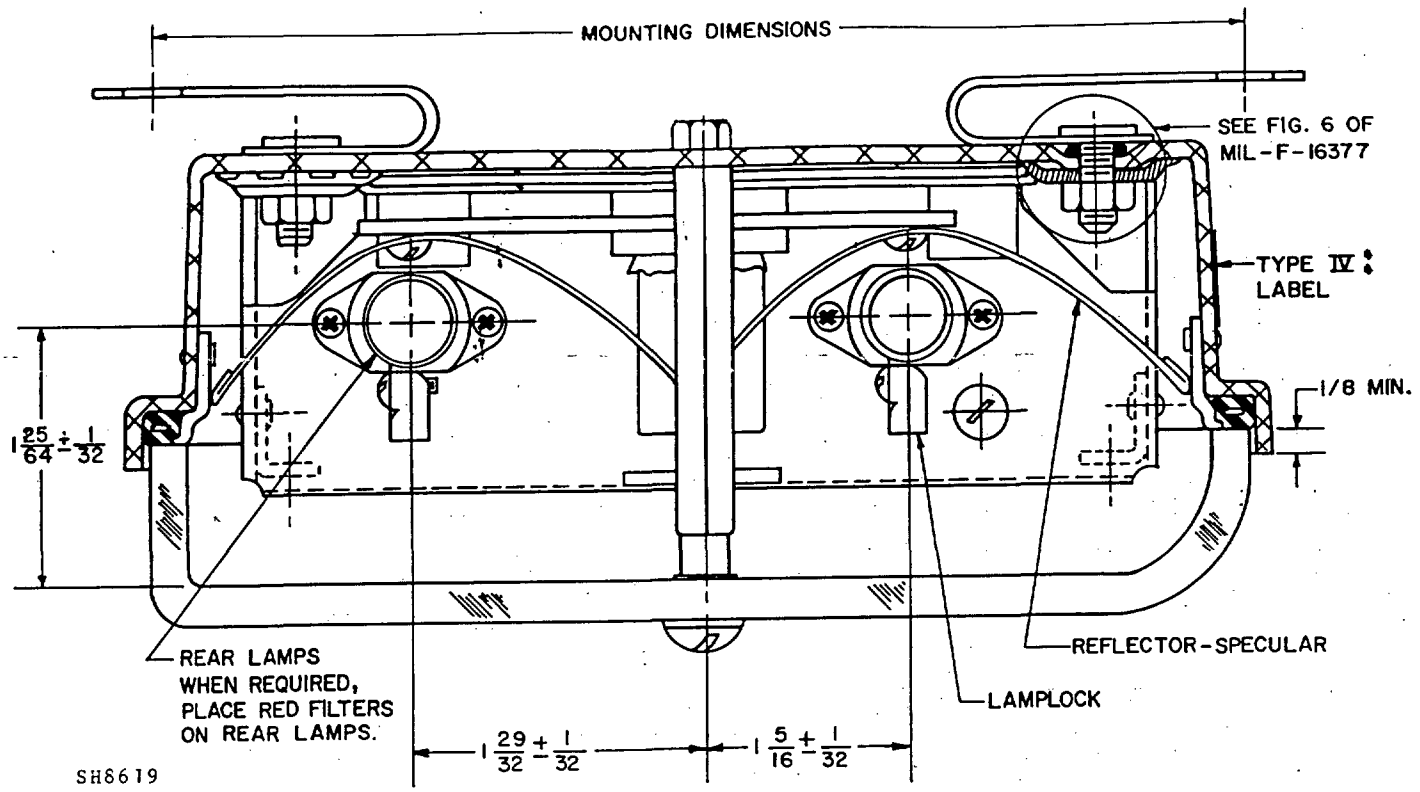



FIGURE 2. Typical section for Symbol 145.4.

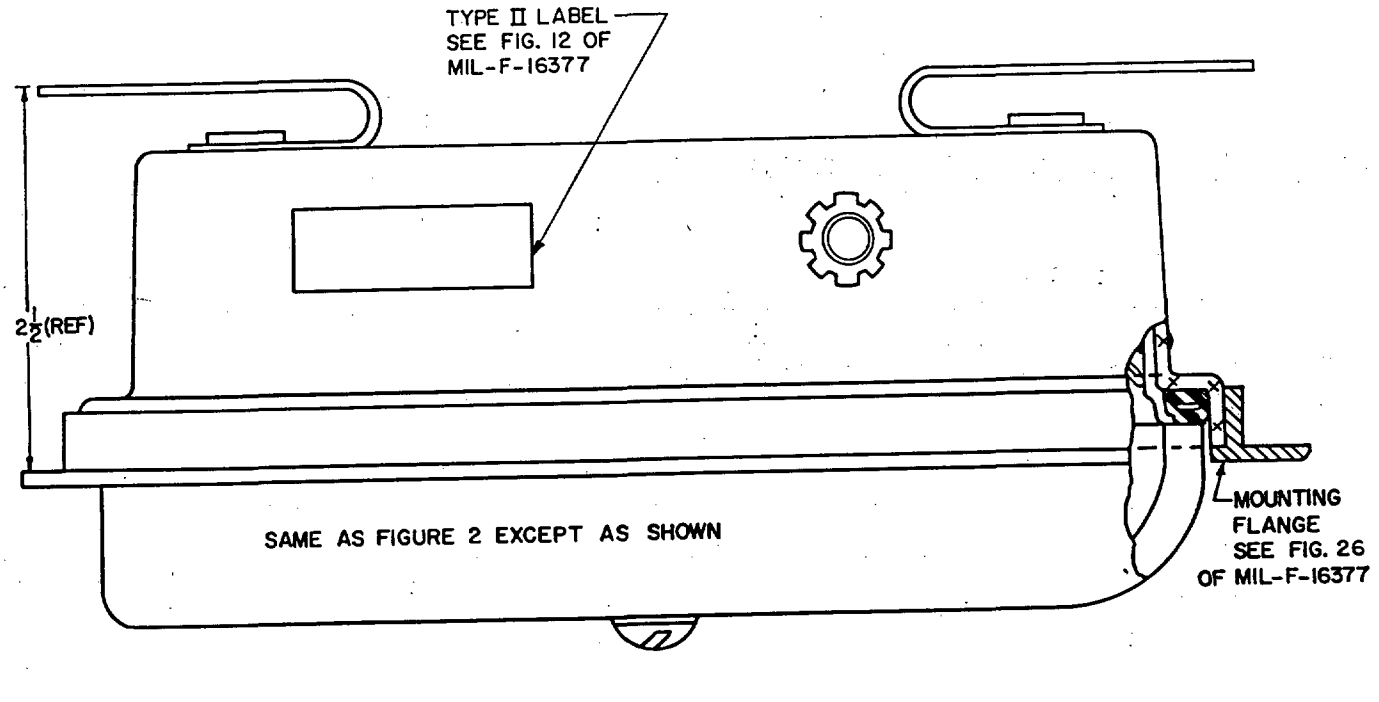
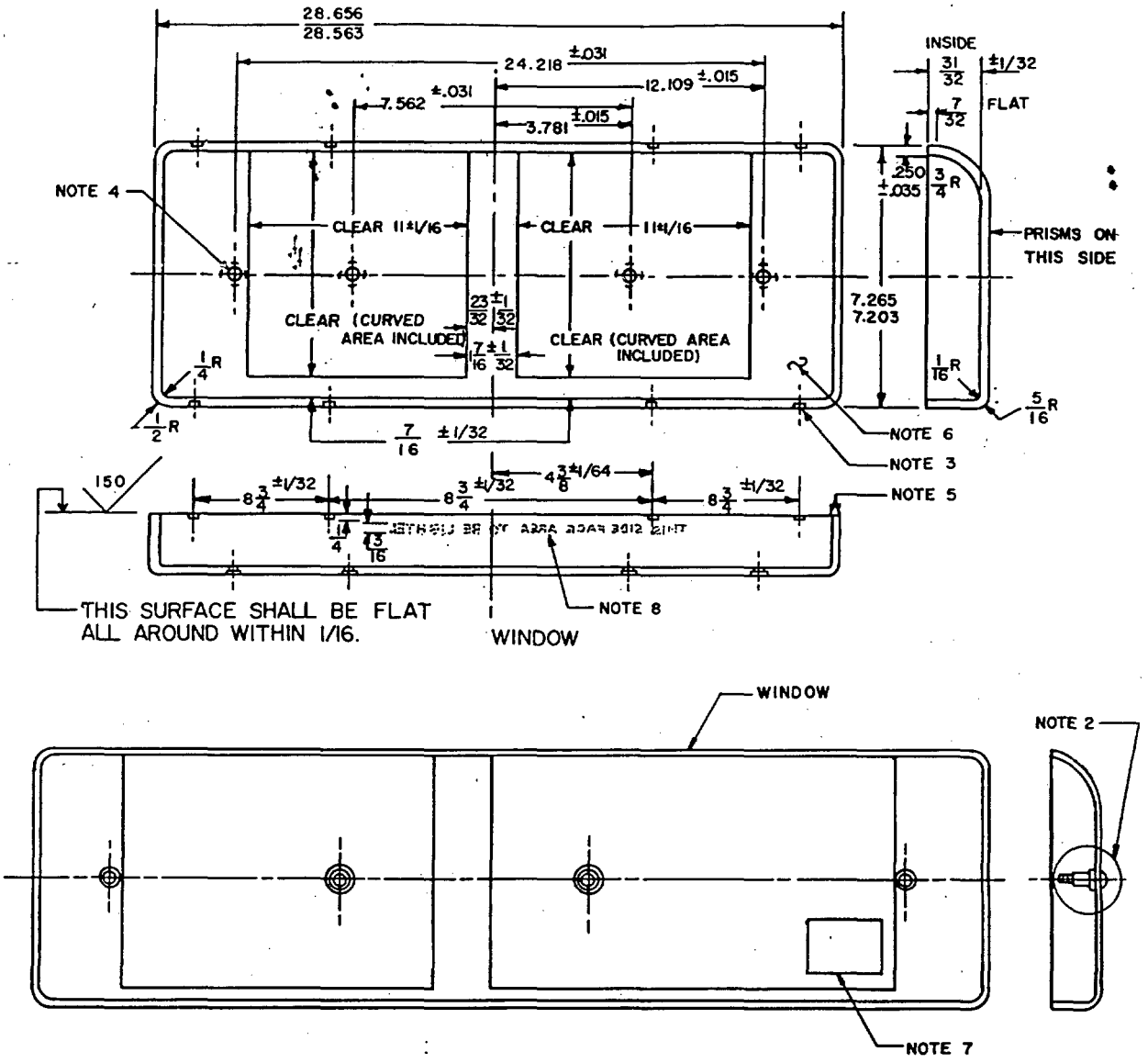



FIGURE 3. Typical section for Symbol 145.5.



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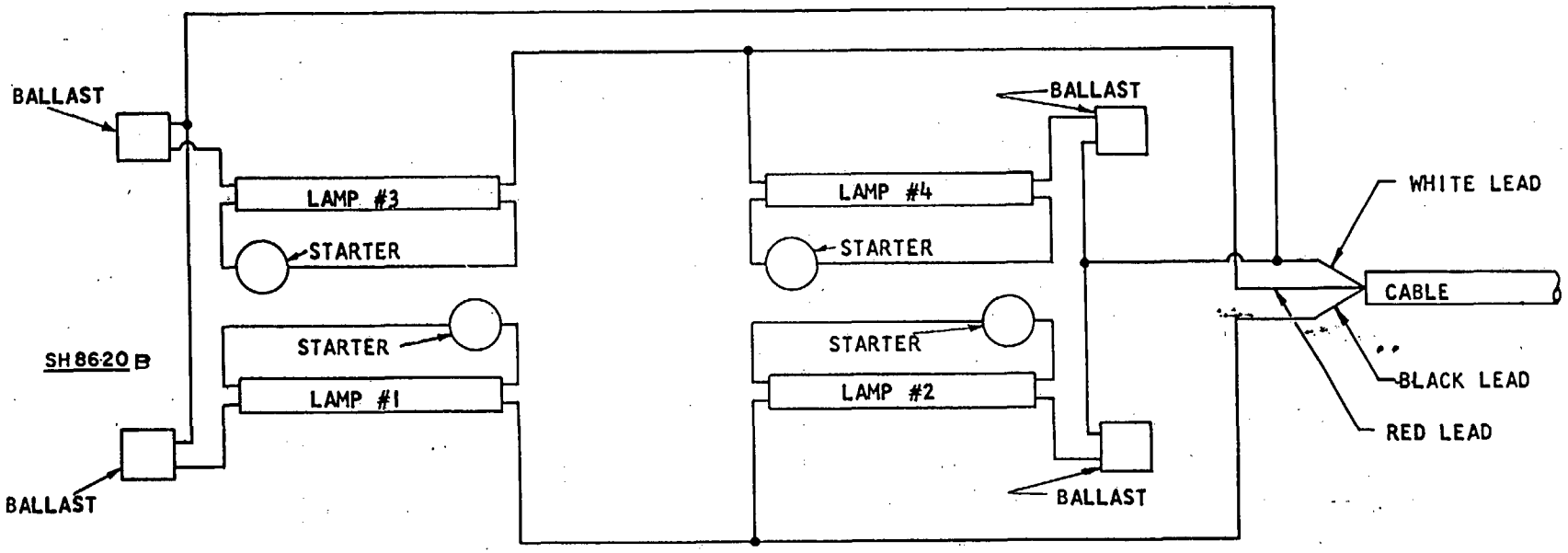
WINDOW ASSEMBLY

1. Material for the window shall be 1/4 inch thick, clear prismatic acrylic plastic, Type CLEAR-2 of MIL-P-24191. The prisms shall be on the outside surface of the windows.
2. Four short window securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-F-16377.
3. When required, eight notches (4 each side) shall be filed as shown in accordance with 3.7.4.4 of MIL-F-16377.
4. Drill 0.191 diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface of all holes.
5. Break sharp edges 1/64 of an inch around entire window.
6. The inside surface of the window except areas indicated as CLEAR shall be finished white in accordance with 3.8.4.1 of MIL-F-16377.
7. When required, one Type VI label shall be applied on the inside surface of the window in accordance with 3.7.4.3 of MIL-F-16377.
8. "THIS SIDE FACES AREA TO BE LIGHTED" shall be stenciled on the inside surface of the curved area readable from the outside surface.

FIGURE 4. Window assembly, clear and white prismatic - directional.



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NOTE: LAMPS #3 AND #4 SHALL BE THOSE DESIGNATED AS REAR LAMPS IN FIGURE 2.

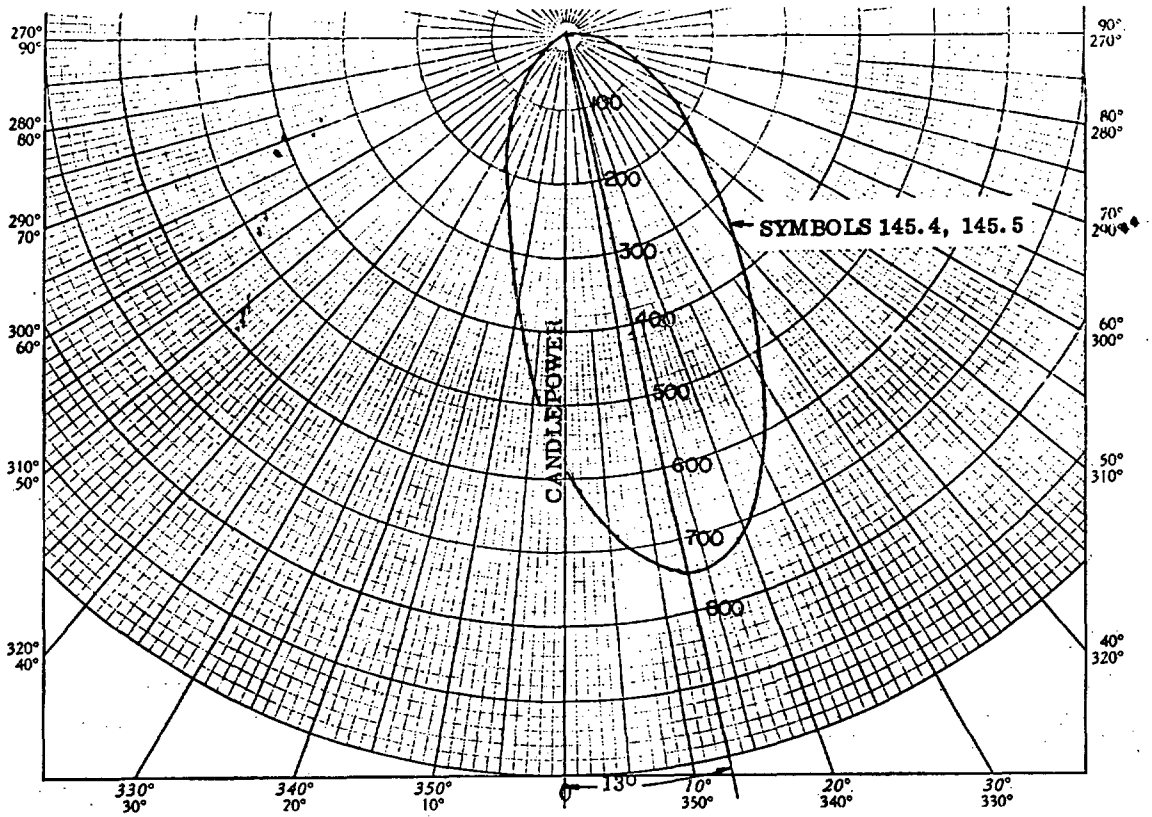
SH8620

FIGURE 5. Wiring diagram.





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| *ANGLE | CANDLEPOWER SYM 145.4, 145.5 |
|---|---|
| 0 | 740 |
| 5 | 720 |
| 10 | 650 |
| 15 | 560 |
| 20 | 450 |
| 25 | 360 |
| 30 | 290 |
| 35 | 240 |
| 40 | 200 |
| 45 | 160 |
| 50 | 120 |
| 55 | 108 |
| 60 | 80 |

| *ANGLE | CANDLEPOWER SYM 145.4, 145.5 |
|---|---|
| 65 | 40 |
| 70 | 20 |
| 75 | 10 |
| 80 | 0 |
| 85 | 0 |
| 90 | 0 |
| 95 | 0 |
| 100 | 0 |
| 105 | 0 |
| 110 | 0 |
| 115 | 0 |
| 120 | 0 |
| 125 | 0 |

* Measured at an angle of 13° from vertical.

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 380 lumens per lamp.

FIGURE 6. Candlepower distribution curves.



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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
145.4: Complete lighting fixture assembly, standard mounting with four 8-watt cool white fluorescent lamps and clear and white prismatic directional window assembly.
145.5: Complete lighting fixture assembly, flush mounting, with four 8-watt cool white fluorescent lamps and clear and white prismatic directional window assembly.

DESIGN:
Electrical rating: 32 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2 and 3 and figure 10 of MIL-F-16377.
Photometric data: see figure 6.
Lampholder spacing: see figure 21 of MIL-F-16377.
Wiring diagram: see figure 5.
Mounting dimensions:
Symbol 145.4: see figure 25 of MIL-F-16377.
Symbol 145.5: see figure 26 of MIL-F-16377.
Window assembly: see figure 4.
Enclosure: Watertight. See 3.5.11 of MIL-F-16377.
Cable: 10 feet of type SJO or SJT, construction in accordance with J-C-580 except AWG size 16, three conductor (white, black, and red).
Weight:
Symbol 145.4 : 12 lbs. 0 oz. maximum.
Symbol 145.5 : 13 lbs. 13 oz. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) Two type I
(b) One type II for Symbol 145.5 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.
(d) One type IV applied as shown in figure 2, bearing legend "THIS SIDE FACES AREA TO BE LIGHTED."

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II
Window assemblies: See table III
Quality conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: MIL-F-16377
Quality conformance tests: See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.



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TABLE I. Standard parts.

| Name | Specification or Military part Number | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| Lamp | W-L-116/3 | 6240-00-299-5546 | Industry Number F8T5/CW |
| Lamplock | M16377/40-001 | 6250-00-399-7039 | |
| Lampholder | MIL-L-970/10 | 6250-00-377-8590 | |
| Startersocket | MIL-L-970/13 | 6250-00-295-2738 | |
| Starter | 8W, type III of W-S-755 | 6250-00-299-5962 | Commercial Designation:FS5 |
| Ballast | M16377/45-001 | 6250-00-295-2735 | |
| Shockmount | M16377/41-009 or M16377/41-010 | 6210-00-973-1747<br>6210-00-973-1748 | See 3.5.16 of MIL-F-16377 |
| Red filter tube | M16377/42-001 | 6210-00-399-6297 | Not furnished See 3.7.7 of MIL-F-16377 |

TABLE II. First article and comparison inspection for complete fixtures.

| Tests | Requirement paragraph | Test paragraph | Remarks |
|---|---|---|---|
| Examination | 3.13 | 4.5 | |
| Operation | 3.13 | 4.8.1 | |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | |
| Insulation resistance | 3.6.16 | 4.8.3 | |
| Enclosure effectiveness | 3.5.11 | 4.8.14.4 | Watertight |
| Shock | 3.5.12 | 4.8.7 | |
| Vibration | 3.5.13 | 4.8.8 | |
| Enclosure effectiveness | 3.5.11 | 4.8.14.4 | Watertight |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | |
| Insulation resistance | 3.6.16 | 4.8.3 | |
| Noise | 3.5.14 | 4.8.9 | |
| Continuity of grounding | 3.6.13.4 | 4.8.12 | |
| Optical uniformity | 3.7.4.1.1 | 4.8.4 | |
| Stress relief | 3.8.5.3 | 4.8.5 | |
| Light output | 3.7.1 | 4.8.6.1 | Candlepower |
| Magnetic permeability | 3.4.8 | 4.8.18 | |



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TABLE III. First article and comparison inspection for window assemblies.

| Tests | Requirement paragraph | Test paragraph | Remarks |
|---|---|---|---|
| Examination | 3.13 | 4.5 | |
| Optical uniformity | 3.7.4.1.1 | 4.8.4 | |
| Stress relief | 3.8.5.3 | 4.8.5 | |
| Shock | 3.5.12 | 4.8.7 | |
| Vibration | 3.5.13 | 4.8.8 | |
| Enclosure effectiveness | 3.5.11 | 4.8.14.4 | Watertight |
| Light output | 3.7.1 | 4.8.6.1 | Candlepower |

TABLE IV. Part number and general information.

| Military Part number 1/ M16377/10- | Name of parts | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| 145.4 | Complete fixture, Symbol 145.4 | 6210-00-548-0227 | New |
| 145.5 | Complete fixture, Symbol 145.5 | | New |
| 002 | Window assembly, clear and white prismatic, directional | | New |

1/ Military Part Numbers M16377/10-145.2, 145.3 and 001 have been superseded by Military part numbers M16377/10-145.4, 145.5 and M16377/67-006 respectively. Superseded fixtures were previously covered by revision -- of this specification, MS17180 and NAVSHIPS Drawing 9000-S6401-74457.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6210-N558)

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